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Unified Neural Lexical Analysis Via Two-Stage Span Tagging

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ABSTRACT

Lexical analysis is a fundamental task in natural language processing, which involves several subtasks, such as word segmentation (WS), part-of-speech (POS) tagging, and named entity recognition (NER). Recent works have shown that taking advantage of relatedness between these subtasks can be beneficial. This paper proposes a unified neural framework to address these subtasks simultaneously. Apart from the sequence tagging paradigm, the proposed method tackles the multitask lexical analysis via two-stage sequence span classification. Firstly, the model detects the word and named entity boundaries by multi-label classification over character spans in a sentence. Then, the authors assign POS labels and entity labels for words and named entities by multi-class classification, respectively. Furthermore, a Gated Task Transformation (GTT) is proposed to encourage the model to share valuable features between tasks. The performance of the proposed model was evaluated on Chinese and Thai public datasets, demonstrating state-of-the-art results.

1 | Introduction

Lexical analysis is a fundamental and critical task in natural language processing. It is composed of several subtasks, such as word segmentation (WS) [1], part-of-speech (POS) tagging [2] and named entity recognition (NER) [3]. Deep learning lexical analysis models are the dominant approach and have also obtained outstanding performance. The following is a detailed description of these subtasks and examples:

1. Word segmentation (WS): Word segmentation is the process of splitting text into words or phrases. For English text, the operation is relatively simple as the words are separated by spaces. However, for Chinese, Thai and other texts, word segmentation is much more complex because the text is not clearly separated by spaces, and the meaning of words is significantly affected by the context. The English text 'I am a particularly quiet person' can be directly split into 'I' and 'am', 'a', 'particularly', 'quiet', 'person'. The Chinese text "我是个特别安静的人" requires a more complex segmentation algorithm, and the result is "我/是/个/特别/安静/的/人".

2. Part-of-speech (POS) tagging: POS tagging is the lexical analysis of the semantic roles played by words in a sentence or text. POS categories include noun, verb, pronoun, adjective etc. POS tagging helps to understand the grammatical structure and semantic relations of a sentence. In

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the Chinese sentence "我坐公交车去酒店 (I took the bus to the hotel.)", through POS tagging, we can get "我/PRON 坐/VERB公交车/NOUN 去/VERB酒店/NOUN".

3. Named Entity Recognition (NER): NER refers to the identification of entity words with specific meanings in text, such as names of people, places, organisations, proper names etc. NER is an important step in information extraction and text comprehension, which helps to extract key information from text. For example, in the Chinese sentence "北京市银行业协会成立于1999年 (Beijing Banking Association was established in 1999)", "北京市 (Beijin)" is labelled as location and "北京市银行业协会 (Beijing Banking Association)" is labelled as organisation.

There is a high correlation between WS, POS tagging and NER. Correct word segmentation is the basis for POS tagging, it is only accurate segmentation that ensures correct POS tagging. Entity boundaries are represented by word boundaries. Entities are composed of one or more words, therefore and entity boundaries must be word boundaries. For the classification of entity tags, part-of-speech labels can also be a useful source of information. As entities are generally nouns or noun phrases. Many researchers have implemented lexical analysis tasks using a multi-task learning strategy to better take advantage of the relationship between them.

In previous studies, the pipeline model was the traditional multi-task learning strategy for lexical analysis tasks. These methods operated independently of each other, resulting in data that cannot be shared across tasks. Tasks are also carried out in a sequential manner, which may cause errors to spread. Specifically, the outcomes of subsequent tasks will be significantly impacted by prediction errors in the previous work. In order to solve the above problems, joint models have been proposed in recent years. This strategy aims to combine many tasks into one unified model for processing in parallel. An encoder is used for joint encoding, on top of which decoding is performed to obtain the final predictions.

According to the different decoding methods, there are generally two types.

1. Sequence tagging models. These methods typically predict the possible states corresponding to each position in the sequence using the CRF decoder. Shao et al. [4] were the first to apply the Bi-RNN-CRF model for general sequence labelling to the joint WS and POS tagging of Chinese, and use the CRF decoder to predict the optimum labelling sequence for the whole phrase.
2. Transition-based models. This framework consists of two parts: states and actions, where actions are used to regulate the transfer between states while states are used to record imperfect prediction outcomes. Yang et al. [5] employed a transition-based approach to simultaneously perform POS tagging and dependency parsing. Their approach utilised a five-action transition system and involved the development of three classifiers to address conflicts related to structure, tagging, and labelling. Zhang et al. [6] presented a transition-based model for joint WS and POS tagging of Chinese. Their model leveraged a transition system that translated the decoding process into the prediction of a series of transition actions.

Despite the significant performance achieved by current multi-task lexical analysis methods, they continue to face two key challenges.

1. In terms of task modelling, previous models usually employed sequence labelling approaches to annotate each character in text individually. However, in complex scenarios of lexical analysis, in order to more accurately delineate words, we need to comprehensively consider contextual information and select the optimal span result by comparing different span combinations.
2. In terms of interaction, existing approaches interact information between tasks via a shared encoder. This approach cannot share and interact with the features deeply, and the correlations between WS, POS tagging and NER tasks are not well explored. In addition, it is unable to perform targeted task interactions such as word and entity boundary interactions and lexical and entity tagging interactions. The performance of multi-task lexical analysis is still limited.

Considering the analysis presented above, we propose a unified framework for two-stage span tagging called TSTLA (Two-stage Span Tagging Lexical Analysis), which models the task as span detection and tag classification using a unified two-stage classification approach. Where span detection includes word and entity boundary detection and tag classification includes POS and entity tag detection. Taking the Named Entity Recognition (NER) task as an example, for the Chinese text "坐火车去北京", traditional sequence labelling methods would assign a label to each character as follows: "坐/O 火/O 车/O 去/O 北/BLOC京/I-LOC". However, the method we propose would first detect the segment "北京" as an entity and then label "北京" with the LOC tag. This approach converts word and entity boundary detection into a multi-label classification problem that often has the problem of category imbalance, so we borrow the idea of circle loss [7] and apply the multi-label classification loss function to the detection layer. Turning the detection process into a two-by-two comparison of span category scores and non-span category scores alleviates the problem well while getting comparable results in terms of performance to CRF. For the possible ambiguity problem in the word separation process, we utilise a straightforward greedy decoding algorithm to get the predicted optimal labels. In addition, we propose the Gated Task Transformation (GTT) module, which is a simple and effective approach. To enhance the information exchange between tasks, we incorporate trainable parameters to encourage the sharing of useful features. The experimental results showcase that our TSTLA model achieves competitive performance on Chinese datasets such as PKU and UD2, as well as the Thai dataset LST20, in comparison to state-of-the-art methods.

The main contributions of this paper can be summarised as follows:

1. We propose a joint model TSTLA for lexical analysis via two-stage span tagging. We consider word/entity detection

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as a multi-class classification task, while POS/entity tagging is also treated as a multi-class classification task. And we achieve the integration of WS, POS tagging, and NER within a unified model framework.

2. We propose a novel information interaction strategy called GTT. This method uses trainable parameters to model the weights of each character. Leverages the correlation between tasks by facilitating the sharing of useful information between tasks.
3. The experimental results provide evidence of substantial improvements achieved by our TSTLA model when compared to the current state-of-the-art method on the PKU, LST20, and UD2 benchmark datasets.

2 | Related Work

2.1 | Lexical Analysis

In the realm of deep learning methods, lexical analysis tasks are commonly reformulated as a sequence annotation problem. The most representative model is BiLSTM-CRF [8], which leverages a BiLSTM as the encoder to extract features. It assigns scores to the predicted labels for each character and employs a Conditional Random Field (CRF) as the decoder to refine the output scores and generate the optimal label sequence. This category of methods inherits the advantages of deep learning approaches, eliminating the need for feature engineering and enhancing the model's feature extraction capabilities. However, they often struggle with recognising unrecognised words. Based on this, the BERT-CRF model is proposed. BERT [9] is an advanced pretrained language model, which can be used as an encoder to produce high-quality word vectors and can effectively alleviate the problem of unrecognised words effectively. However, the training cost and complexity of using CRF for decoding are high. The NER task may also contain flat and nested entities. Sequence tagging methods cannot directly identify all nested entities, thus have obvious limitations. To address this issue, Yu et al. [10] proposed a two-stage classification approach. The task of entity extraction is treated as the problem of recognising start and end indices. This model proposes to use biaffine at the top of a multilayer neural network to output the scores of different entity types for all spans, rank the candidate spans based on these scores, and then predict the entity type of the spans by the highest score. In addition, Zhang and Yang [11] introduced the lattice LSTM model, which integrates character information with lexical information for the first time. Compared to character-based methods, the lexical-integrated methods obviously incorporate more information. However, in contrast to vocabulary-based methods, lattice LSTM avoids the performance degradation caused by the impact of word segmentation errors on LSTM.

In recent years, Large Language Models [12] (LLMs) have been widely applied to natural language processing tasks such as text generation, sentiment analysis, and question-answering systems achieving remarkable results. These LLMs are trained and fine-tuned on vast amounts of text data, gaining a deep understanding of language. Wang et al. [13] proposed a method called GPT-NER, which converts a NER task into a text generation task, aiming to get comparable performance to supervised learning. Kim et al. [14] investigated the application of GPT-2 and BERT for named entity disambiguation in dialogue systems and verified the effectiveness of this combined strategy. Experimental results show that the approach using GPT-2 is slightly less accurate compared to BERT. Dukic et al. [15] investigated the capability of decoder-only LLMs for sequence labelling. They employed a technique to enhance the sequence labelling performance of open LLMs on information extraction tasks by applying a layered removal of causal masking during the fine-tuning process of the LLMs. Keraghel et al. [16] have conducted a comprehensive review of the latest advancements in NER, including the recently highly popular LLMs. The analysis reveals that, despite their excellent language understanding capabilities, large language models do not perform prominently in NER tasks. The performance of LLMs on lexical analysis tasks is less satisfactory, primarily because natural language understanding tasks such as text generation, sentiment analysis, and question-answering systems primarily rely on the models' overall understanding and generative capabilities of language. By comprehending semantics, LLMs can produce coherent and contextually appropriate text or answers to users' questions. In contrast, lexical analysis is a more fundamental and intricate natural language processing task, requiring accurate identification and analysis of word morphology and grammatical functions. This demands that the model possess an in-depth understanding and fine-grained processing ability of language structures, which is relatively weak in LLMs.

2.2 | Multi-Task Learning for Lexical Analysis

Because of its strong association with lexical analysis, the multi-task learning strategy was adopted and proved its effectiveness. There are two main approaches to multi-task learning for lexical analysis: the pipeline model [17, 18] and the joint model. Previous studies have shown that the joint models outperform pipeline models. The existing joint learning methods can be categorised into two main types: cascaded sequence labelling methods and transition-based methods.

2.2.1 | Pipeline Models

For example, word segmentation is performed first, and then the results of the previous stage are used to tag the POS labels. Such methods increase the overall complexity of the model. Not only is it prone to error propagation, but also the tasks cannot use their associativity to promote each other.

2.2.2 | Joint Models

The joint model aims to mitigate the issue of error propagation in pipeline models by integrating multiple tasks into a single end-to-end model.

1. Sequence Tagging Methods. Sequence tagging is a sequence-to-sequence process that involves assigning labels to each position of an input sequence using a model.

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In joint tasks, data sharing is usually performed using a shared encoder while predicting tags using a CRF-based sequence tagging framework. Shao et al. [4] utilise the BiRNN-CRF model for sequence tagging in the context of joint WS and POS tagging. Their approach is to predict combination tags that include both word boundaries and POS tags. Concatenated n-grams and lower-than-character features are applied along with conventional embeddings as vector representations for Chinese characters. Jiao et al. [19] proposed a method for a combined lexical analysis task involving word segmentation, POS tagging, and NER. They introduced a stacked Bi-GRU neural network with a CRF decoding layer to address this task effectively. To train their model, they leveraged multiple extensive corpora that were pre-tagged using their most advanced Chinese lexical analysis tools. Additionally, they incorporated a small, yet meticulously annotated corpus of high quality to further enhance the performance of model. In summary, despite the remarkable performance demonstrated by the sequence labelling method in lexical analysis tasks, its approach of annotating each character may result in compromised performance when confronted with texts that exhibit rich morphological variations or complex syntactic structures.

2. Transition-based Methods. The transition-based framework has found extensive application in various structural learning problems, enjoying widespread adoption. It consists of two parts: states and actions. The states are used to record incomplete predictions and the actions are used to control the transfer between states. Zhang et al. [20] conducted research on Chinese parsing from the character level, extending the concept of phrase structure trees by annotating the internal structures of words, and built an integrated system for joint WS, POS tagging, and phrase-structure parsing. Lyu et al. [21] conducted research on character-level Chinese chunking based on the transition-based framework. To address the issue of chunk features, they employed a semi-supervised approach, leveraging large-scale automatically chunked text data, which effectively improved the accuracy of segmentation and chunking. Yang et al. [5] propose an innovative method that expands upon a transition-based model for dependency parsing, enabling joint treatment of POS tagging and dependency parsing. By employing a five-action transition system, they develop three classifiers specifically designed to resolve conflicts related to structural dependencies, POS tagging, and labelling. This approach offers a comprehensive solution for effectively integrating POS tagging and dependency parsing tasks. Zhang et al. [6] introduce a Seq2Seq model designed for joint WS and POS tagging. This model is constructed within a framework that utilises a transition system to transform the decoding process of the joint task into predicting a sequence of transition actions. This neural model incorporates two key components, both based on LSTM networks. The first component is a Bi-LSTM that operates on the input sequence, whereas the second component comprises a left-to-right LSTM that directly processes historical word-tag pairs acquired from the transition system. These components form the foundation of the neural model, enabling effective integration of WS and POS tagging in a unified approach. The method treats the lexical analysis as a single-label classification task, so it is therefore prone to overfitting with limited training data.

3. Span Labelling-Based Methods. These approaches redefine the lexical analysis task as the identification of start and end indices, along with assigning a corresponding category to the span encompassed by these pairs. Dozat et al. [22] first proposed the idea to the dependency parsing, using biaffine classifiers to predict arcs and labels. Yu et al. [10] applied biaffine classifiers for migration to the NER task, dividing the task into span detection and label classification, which can identify nested entities more efficiently. Nguyen et al. [23] introduce a neural model to handle the task of Chinese WS and POS tagging simultaneously, employing a span labelling approach. In this approach, the primary challenge lies in determining the probability of each n-gram serving as a word and its corresponding POS tag. To model the n-grams and POS tags, the researchers utilise a biaffine operation, which operates on the left and right boundary representations of consecutive characters. This operation enables effective modelling of the relationships between the n-grams and their associated POS tags. These methods are more intuitive for humans, but they lead to problems with category imbalance.

In addition, in order to effectively utilise the correlation between subtasks, Chen and Qian [24] proposed a RelationAware Collaborative Learning (RACL) framework which supports three subtasks, term extraction, opinion term extraction, and aspect-level sentiment classification. Collaborative work is achieved through multi-task learning and relationship propagation mechanisms in stacked multi-layer networks. Kang et al. [25] proposed a Multi-Task Learning model for text normalisation, POS tagging, and homograph disambiguation. The framework adopts a tree structure, where the trunk learns shared representations. Furthermore, it integrates a pre-trained language model to leverage its built-in vocabulary and contextual knowledge. We aim to propose a lightweight method that, compared to previous models, can utilise the interdependencies between tasks in a simpler and more efficient manner.

3 | Method

Figure 1 illustrates the architecture of the TSTLA model. The input is encoded by using a pre-trained language model (PLM) within the model, which generates feature vectors that are then fed into the GTT layer. The GTT layer facilitates the interaction of information between each task, enabling effective task-specific learning. Taking Named Entity Recognition as an example, the GTT layer incorporates the entity tag classifier task as the primary task and the entity classifier task as the auxiliary task. Furthermore, the TSTLA model adopts a two-stage classification approach. The initial stage revolves around detecting word/entity boundaries, whereas the subsequent stage centres on classifying POS/entity tags. This two-stage approach enables accurate identification of entity boundaries and appropriate assignment of entity tags within the NER task.

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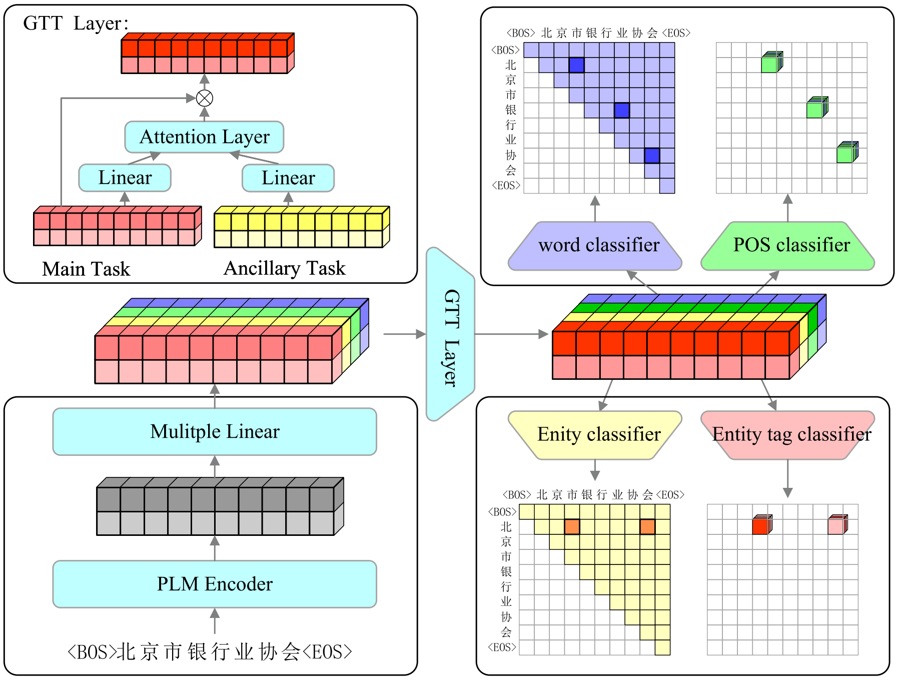

FIGURE 1 | The structure of the TSTLA model.

3.1 | Encoder

We use a PLM to encode each sentence and using characters as input units. The input sentence is represented as a sequence of characters, denoted as $w = \{w_1, w_2, w_n\}$, where n represents the number of the character. Each character w_i corresponds to a specific token in the BERT vocabulary.

The sequence w is fed into the PLM to get the vector representation $\boldsymbol{h} = [\boldsymbol{h}_1, \boldsymbol{h}_2, \boldsymbol{h}_n]$, where $\boldsymbol{h}_i \in R^d$ is the vector representation corresponding to w_i, vector dimension $d = 768$.

The encoded vector sequence $\boldsymbol{h}$ is transformed $\boldsymbol{q}_i^{(t)} = W_q^{(t)}\boldsymbol{h}_i^{(t)} + b_q^{(t)}$ and $\boldsymbol{k}_i^{(t)} = W_k^{(t)}\boldsymbol{h}_i + b_k^{(t)}$ to get the vector sequences for each task $[\boldsymbol{q}_1^{(t)}, \boldsymbol{q}_2^{(t)}, \boldsymbol{q}_n^{(t)}]$ and $[\boldsymbol{k}_1^{(t)}, \boldsymbol{k}_2^{(t)}, \boldsymbol{k}_n^{(t)}]$, where $\boldsymbol{q}_i^{(t)} \in R^{n\times d'}$, $\boldsymbol{k}_i^{(t)} \in R^{n\times d'}$, T stands for each task, $W_q^{(t)}$ and $W_k^{(t)}$ are the parameters of the model.

3.2 | Gated Task Transformation

Gated task transformation is to set highly associated tasks as auxiliary tasks and interact their valuable features with those of the main task. Full information interaction is performed by sharing useful features between tasks. Task features include $\boldsymbol{q}_i^{(t)}$ and $\boldsymbol{k}_i^{(t)}$, which are analysed in the following as an example of $\boldsymbol{q}_i^{(t)}$.

A linear transformation is applied to downscale $\boldsymbol{q}_i^{(t)}$, which is used to aggregate the feature information of the share task. The definition is as follows Equation (1) and Equation (2):

$$\boldsymbol{v}_i^{(t)} = \tanh\left(\boldsymbol{W}^{(t)}\boldsymbol{q}_i^{(t)} + b^{(t)}\right) \tag{1}$$

$$\boldsymbol{v}_i^{(t')} = \tanh\left(\boldsymbol{W}^{(t')}\boldsymbol{q}_i^{(t')} + b^{(t')}\right) \tag{2}$$

where $\boldsymbol{W}^{(t)}$ and $\boldsymbol{W}^{(t')}$ represent the weight parameters associated with the auxiliary task and the primary task, respectively. $\boldsymbol{q}_i^{(t)}$ and $\boldsymbol{q}_i^{(t')}$ are the initial features of the auxiliary task and the primary task. $\boldsymbol{v}_i^{(t)}, \boldsymbol{v}_i^{(t')} \in R^{n\times d''}$ and $\boldsymbol{q}_i^{(t)}, \boldsymbol{q}_i^{(t')} \in R^{n\times d'}$.

It captures the relationship between the features of the auxiliary task$\boldsymbol{v}_i^{(t')}$ and the main task $\boldsymbol{v}_i^{(t)}$ through linear transformations and attention mechanisms, thereby modelling their correlation.

It functions to establish a competitive relationship between tasks. It makes the values of features that have a large impact on the main task relatively large, suppressing other features that have smaller feedback. The definition is as follows Equation (3):

$$s^{(t',t)} = \boldsymbol{v}_i^{(t')}\boldsymbol{v}_i^{(t)\top} \tag{3}$$

where $s^{(t',t)} \in R$.

The calculated ownership weights are aggregated through the activation function, denoted in Equation (4) as $\boldsymbol{a}$:

$$a^{(t)} = \mathrm{softmax}\left(s^{(0,t)}, s^{(1,t)}, \ldots, s^{(t',t)}, \ldots, s^{(T,t)}\right) \tag{4}$$

where $a^{(t)} \in R^T$.

By introducing a gating mechanism, it can adjust the character of each word and promote competitive and synergistic

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relationships between tasks. The definitions are as follows Equation (5):

$$\hat{\boldsymbol{q}}_{\boldsymbol{i}}(\boldsymbol{t}) = a^t \boldsymbol{q}_i^{(t)} \tag{5}$$

where $\hat{\boldsymbol{q}}_{\boldsymbol{i}}(\boldsymbol{t}) \in R^{n\times d'}$.

The calculation process of $\hat{\boldsymbol{k}}_{\boldsymbol{i}}(\boldsymbol{t})$ is consistent with $\hat{\boldsymbol{q}}_{\boldsymbol{i}}(\boldsymbol{t})$. The features of each task after the interactions are $\hat{\boldsymbol{q}}(\boldsymbol{t}) = [\hat{\boldsymbol{q}}_{\boldsymbol{0}}(\boldsymbol{t}), \hat{\boldsymbol{q}}_{\boldsymbol{1}}(\boldsymbol{t}), ..., \hat{\boldsymbol{q}}_{\boldsymbol{n}}(\boldsymbol{t})]$ and $\hat{\boldsymbol{k}}(\boldsymbol{t}) = [\hat{\boldsymbol{k}}_{\boldsymbol{0}}(\boldsymbol{t}), \hat{\boldsymbol{k}}_{\boldsymbol{1}}(\boldsymbol{t}), ..., \hat{\boldsymbol{k}}_{\boldsymbol{n}}(\boldsymbol{t})]$.

3.3 | Word and Entity Boundary Detection

We detect words and entities by classify consecutive spans in input sentence. Specifically, the input of length n has $n(n+1)/2$ different consecutive subsequences. Without considering POS and entity categories, all words and entities are selected from the $n(n+1)/2$ consecutive subsequences.

Take the example of the word boundary detection, calculate the word score $\boldsymbol{s}_{(\boldsymbol{i},\boldsymbol{j})}{}^{(\mathrm{ws})} \in R$ for a continuous subsequence $w_{[i:j]}$ by Equation (6):

$$s_{(i,j)}^{(\mathrm{ws})} = \hat{\boldsymbol{q}}_i(\mathrm{ws})^{\top} \hat{\boldsymbol{k}}_j(\mathrm{ws}) \tag{6}$$

where (i, j) is a continuous span from i to j.

Entity boundary detection is the same process as described above for word boundary detection to obtain $s_{(i,j)}^{(\mathrm{ne})}$.

3.4 | Part-Of-Speech and Entity Tag Classification

The filtered words and entities are classified by labelling. Take the example of the POS tag classification, the $\hat{h}$ after GTT are predicted POS labels given as follows:

$$s_{(i,j)}^{(\mathrm{pos})} = \hat{\boldsymbol{q}}_i(\mathrm{pos})^{\top} U \hat{\boldsymbol{k}}_j(\mathrm{pos}) \tag{7}$$

where $\boldsymbol{s}_{(i,j)}^{(\mathrm{pos})} \in R$, $U \in R^{l\times l}$, l is the number of labels, $\hat{\boldsymbol{q}}_i(\mathrm{pos})^{\top} U \hat{\boldsymbol{k}}_j(\mathrm{pos})$ is the label probability in the case, i is the head and j is the tail.

The entity tag classification is consistent with the part-of-speech tag classification process described above, yielding $s_{(i,j)}^{(\mathrm{ner})}$.

3.5 | Loss Function

Our proposed model has four loss functions, which include the WS, entity boundary, POS tagging and NER loss function. The calculation method for WS and entity boundary loss function is similar. Taking the WS loss function as an example, it is represented by Equation (8). Similarly, POS tagging and NER loss function are calculated in a similar manner. For instance, the word segmentation loss function is demonstrated by Equation (9).

$$\mathcal{L}_{\mathrm{span}}^{\mathrm{ws}} = \log\left(1 + \sum_{(i,j)\in\mathcal{P}} e^{-s_{(i,j)}^{\mathrm{ws}}}\right) + \log\left(1 + \sum_{(i,j)\in\mathcal{Q}} e^{s_{(i,j)}^{\mathrm{ws}}}\right) \tag{8}$$

where $s_{(i,j)}$ is the score in Section 2.2.2, $\mathcal{P}$ is the set of word/entity start and end for that sample, and $\mathcal{Q}$ is the set of all non-word/non-entity start and end for that sample.

$$\mathcal{L}_{\mathrm{cls}}^{\mathrm{ner}} = \frac{1}{|P|} \sum_{(i,j)\in\mathcal{P}} \left(-\sum_{c=1}^{|C|} y_{(i,j)}^{c} \hat{y}_{(i,j)} c \right) \tag{9}$$

where C is the set of label categories, $y_{(i,j)}^{c} \in [0, 1]$ is the label of $x_{[i:j]}$ on category C, and $\hat{y}_{(i,j)} c \in [0, 1]$ is the predicted value of the model on category c.

Finally, the total loss function of TSTLA *Loss* is the sum of four loss functions by Equation (10):

$$\mathcal{L} = \mathcal{L}_{\mathrm{span}}^{\mathrm{ws}} + \mathcal{L}_{\mathrm{span}}^{\mathrm{ne}} + \mathcal{L}_{\mathrm{cls}}^{\mathrm{pos}} + \mathcal{L}_{\mathrm{cls}}^{\mathrm{ner}} \tag{10}$$

3.6 | Word Segmentation Inference Algorithm

In the word segmentation task, identifying word boundaries by detecting spans has the possibility of overlapping spans or word miss-labelling.

During the detection process, for the characters with missing labels, we all predict it as a single character as a word. We use a greedy algorithm to get the best solution to ensure that there is no overlap between the two spans. The predicted span $\hat{s}$ is formalised as follows:

ALGORITHM 1 | Greedy word segment algorithm.

Input: $S \in \mathbb{R}^{n\times n}$ is the score triangular matrix of word spans, n is the length of input sentence; tokens: a string list generated by PLM tokeniser
Output: $\mathcal{W}$: list of the predicted words initialise $i = 0, j = 0$ and $\mathcal{W} = \phi$

```
while i < n do
    j = argmax(s[i, :])
    W = W ∪ {tokens[i : j + 1]}
    i = j + 1
end while
return W
```

4 | Experiments

In this section, we begin by presenting datasets and evaluation metrics employed in our experiments. We then explain our experimental implementation details, baseline methods and report our experimental results.

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4.1 | Dataset

We use three different datasets for our experiments, namely PKU, LST20 [26] and the Chinese part of Universal Dependencies 2.4 [27] (UD2), where PKU and UD2 are Chinese datasets and LST20 is a Thai dataset. The statistics for the mentioned datasets are summarised in Table 1.

The PKU dataset [1] is created through POS tagging and WS of the plain text corpus of People's Daily from the first half of 1998. It encompasses word segmentation, POS tagging, flat named entities, and nested entities. The dataset comprises 26 POS tags and 5 entity tags.

The LST20 dataset [2] includes five layers of linguistic annotation: word segmentation, part-of-speech tagging, flat named entities, clause boundaries, and sentence boundaries. It encompasses 16 POS tags and 10 entity tags.

The UD dataset [3] is in traditional Chinese, and we follow [4] to convert it to simplified Chinese [4] before conducting experiments on the UD dataset. This includes 15 POS tags.

4.2 | Evaluation

In our experiments, we utilise precision rate (P), recall rate (R), and F1 score as evaluation metrics for word segmentation, following the guidelines set by SIGHAN 2005 [5, 28], which are defined as follows Equation (11)–(13):

$$P = \frac{\#correct}{\#predict} \tag{11}$$

$$R = \frac{\#correct}{\#gold} \tag{12}$$

$$F1 = \frac{2 \times P \times R}{P + R} \tag{13}$$

where #correct is the number of correctly segmented words, #predict is the number of words in prediction and #gold is the number of gold words.

We use the Macro-averaging F1-score and the Micro-averaging F1-score as evaluation criteria in the POS tagging and NER tasks, calculated by not classifying the data set and then calculating the F1 value. Acro-averaging F1 value is averaged for each class F1 value and then for all classes. Micro-averaging F1 value is calculated by not classifying the data set and then calculating the F1 value. Among them, the precision of POS tagging is the ratio between the number of both correctly segmented words and annotated part of speech and the number of POS tagging in prediction; the recall of POS tagging the ratio between the number of correctly segmented words and annotated parts of speech and the number of gold POS tagging. The precision of NER is the ratio between the number of correctly identified entities and the number of entities in prediction; the recall of NER the ratio between the number of correctly entities and the number of gold entities. Our evaluation methods for the POS tagging and NER tasks followed the research of Tian et al. [6, 29].

TABLE 1 | Statistics of datasets.

		PKU	LST20	UD2
Train	# sent	87k	63k	4k
	# word	826k	271k	99k
Dev	# sent	12k	5.6k	500
	# word	118k	241k	13k
Test	# sent	25k	5.2k	500
	# word	237k	207k	12k
# POS tags		46	16	15
# NER tags		3	10	—

4.3 | Implementation Details

In this paper, we utilise the Adam [30] gradient descent algorithm for training the model. To facilitate the training process, we employ a fixed-step learning rate decay strategy, where the decay step occurs at the end of each epoch. The hardware and software environments and the training/testing time are shown in Tables 2 and 3.

The input unit for a Chinese dataset is a character. And in the Thai dataset, we use the syllable tokenisation SSG to make syllables the input unit. Since our model needs to complete the WS task, input granularity is smaller than the words, so we used bert-base-chinese [7] and wangchanberta [8] as PLM on the Chinese and Thai datasets, respectively.

The initial learning rate of the model is set to 5e-6. To accommodate small-scale datasets UD2, we set the initial learning rate to 4e-6. The batch size of the training data is set to 10, the training process is early stopping, and the stopping metric is the maximum POS micro F1 value. The model hyperparameter settings are detailed in Table 4.

4.4 | Baseline Methods

To assess the effectiveness of the TSTLA model, we compare it with several state-of-the-art methods chosen as baselines. On the PKU dataset, we evaluate the TSTLA model against the following methods.

TABLE 2 | Hardware and software environment.

Project	Environment
GPU	NVIDIA GeForce RTX 2080 Ti
Memory	12G
Hard Disk	2T
System	Ubuntu16.04 LTS
Python version	Python 3.8
Pytorch version	1.10.1

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TABLE 3 | Training/testing time of the TSTLA model.

Dataset	Training time	Testing time
PKU	900 min	15 min
LST20	600 min	15 min
UD2	180 min	2 min

TABLE 4 | Hyperparameters used in our experiment.

Parameter	Value
Batch size	20
Dectect hidden size	64
Classify hidden size	64
Optimiser	Adam
Weight decay	1e-3
Learning rate	5e-6

CNN-CNN-LSTM [31], an architecture for NER consists of a character-level encoder using convolutional layers, a word-level encoder using convolutional layers, and an LSTM tag decoder.

BiLSTM-CRF [32], a CRF model utilises features generated by a BiLSTM model, with glove word embeddings employed to convert words into continuous vectors.

BERT-BiLSTM-CRF [32], a BERT-based model that takes contextual representations from BERT and feeds them into a BiLSTM model to obtain hidden representations. These hidden representations are then passed through a CRF layer to generate a sequence of labels.

CRNN-RNN [32], a convolutional neural network is used as an encoder for extracting features. The decoder is a modified recurrent neural network, which is improved by introducing convolutional structure into the recurrent neural network. The gating mechanism uses a GRU cell that introduces the output value of the previous moment softmax classification network into the current hidden layer state while preserving the temporal self-connection of the hidden layer.

Undirected graphical model [33], based on the undirected graph model, deeper dependencies are used as features to unify subword and lexical annotation in a sequential annotation model.

BERT-BiLSTM-CRF-MFM [34], a parallel model designed for WS and POS tagging. It builds upon the BERT-BiLSTM-CRF model and incorporates the Markov family model (improved hidden Markov model) to calculate the transition probability within the CRF layer for WS and POS tag inference.

On the LST20 dataset, we compare the TSTLA model with the following methods.

DeepCut [35], a Thai word separation model, uses multiple CNN filters as feature encoders. It takes characters and their types as input.

AttaCut-C [36], a CNN-based splitting model for Thai language, uses an expanded CNN filter as a feature encoder.

AttaCut-SC [36], a variant model of AttaCut-C, which uses syllable embedding as an additional input.

XLMR (XLM-RoBERTa) [37] and **mBERT (Multilingual BERT)** [9], pre-trained multilingual models based on BERT.

CRF [38], implemented using CRFSuite [39]. By using the Viterbi algorithm to find the best path for the sequence of words corresponding to the sequence of tags.

Wangchanberta-spm-CRF [40], a PLM based on RoBERTa-base was used to apply Thai-specific text processing rules and propose a language model with multiple cut-off granularities for decoding using CRF. Wangchanberta-spm-CRF is based on SentencePiece for sentence slicing and has the best performance of various granularities.

We selected the best performing Wangchanberta-spm-CRF model for comparison experiments.

Wangchanberta-syllable-CRF [40], the only difference between Wangchanberta-syllable-CRF and Wangchanberta-spm-CRF is the difference in granularity. Wangchanberta-syllable-CRF slices the sentences based on syllables.

On the UD2 dataset, we compare the TSTLA model with the following methods.

BiRNN-CRF [4], which is a CRF model that employs features generated by a BiRNN model. This model integrates Chinese segmentation and POS tagging by predicting combination tags for word boundaries and POS tags jointly.

TWASP (BERT) [29] and **TWASP (ZEN)** [29], a character sequence-based annotation model, using an existing language processing toolkit to obtain syntactic knowledge for automatic analysis of the context, and a two-way attention mechanism to combine contextual features and their corresponding syntactic knowledge into the input characters, enabling joint annotation of WS and POS. Particularly, two encoders are used for testing: BERT and ZEN.

SPANSEGTAG [23], which utilises biaffine attention on the left and right boundary representations of consecutive characters to model the n-grams. This model incorporates joint WS and POS labelling through span labelling.

5 | Experimental Results and Discussion

5.1 | Overall Performance Analysis

Table 5 shows the overall performances on the test datasets of PKU, LST20 and UD2. We can see that our TSTLA model achieves competitive results on PKU, LST20 and UD2.

According to the results presented in Table 5, it can be concluded that the TSTLA model surpasses other models in all

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TABLE 5 | Performance comparison of different methods on different dataset.

Dataset	Model	WS F1	POS Micro F1/Macro F1	NER Micro F1/Macro F1
PKU	CNN-CNN-LSTM	—	—	90.81/-
	BiLSTM-CRF	—	—	90.08/91.02
	BERT-BiLSTM-CRF	—	—	94.86/-
	CRNN-RNN	—	—	94.25/94.11
	Undirected graphical model	97.06	95.22/-	—
	BERT-BiLSTM-CRF-MFM	98.03	97.52/-	—
	TSTLA	**98.25**	**97.65/76.33**	**97.11/96.75**
LST20	DeepCut	96.00	—	—
	Attacut-C	93.49	—	—
	Attacut-SC	94.28	—	—
	XLMR	—	96.57/85.00	73.61/68.67
	mBERT	—	96.44/**85.86**	75.05/68.25
	CRF	—	96.28/81.28	75.94/72.13
	Wangchanberta-syllable-CRF	—	96.06/83.98	76.45/70.37
	Wangchanberta-spm-CRF	—	**96.62**/85.44	78.01/72.25
	TSTLA (GOLD)	—	96.41/83.61	**78.86/72.89**
	TSTLA (PRED)	**98.05**	94.88/82.84	78.55/72.62

Note: The bold values are the-state-of-art results.

evaluation metrics for the lexical analysis task on the Chinese PKU dataset. When compared to joint models for WS and POS tagging, such as the undirected graphical model and BERT-BiLSTM-CRF-MFM, TSTLA demonstrates superior performance in both WS and POS tagging, establishing itself as the state-of-the-art approach. This affirms the effectiveness of our proposed joint model. In the NER task, TSTLA showcases significant improvements in both macro-averaged F1 values and micro-averaged F1 values compared1 to models that do not incorporate the BERT pre-trained language model. This indicates the powerful language representation and feature extraction capabilities of the BERT pre-trained language model. Furthermore, when compared to BERT-BiLSTM-CRF, TSTLA achieves optimal performance, further validating the effectiveness of our proposed interaction method, further validating the effectiveness of our proposed interaction method.

The results of our model and previous studies on LST20 datasets are shown in Table 5. Since previous POS tagging experiments in the Thai LST20 dataset were performed with known correct word segmentation, we conducted two experiments: TSTLA (GOLD) and TSTLA (PRED). The former is a joint learning of POS tagging and NER with known correct word segmentation, and the latter is a joint learning of WS, POS tagging and NER with unknown correct word segmentation. Compared to the other models, TSTLA (GOLD) obtained optimal performance on the NER task and comparable performance on the POS task. TSTLA (PRED) had the highest F1 value for WS. The performance of POS tagging for TSTLA (PRED) is slightly lower because TSTLA (PRED) POS labels are tagged under predicted words, whereas the other models are tagged under correct words. Among them, two neural network-based pre-trained language models, XLMR and mBERT, are less effective in the NER task. Although the CRF-based approach models the sequential relationship between labels, the traditional machine learning is less capable of acquiring linguistic features compared to the neural network model. Compared to Wangchanberta-syllable-CRF, we use the same pre-trained language model, with the difference that TSTLA's decoding approach adopts the idea of fragment scoring and does not consider the sequential relationship, making it simpler than the CRF decoding approach, while allowing for better task association. Compared with Wangchanberta-spm-CRF, because the segmentation granularity of this model is SentencePiece, the word information can be better obtained, so the result of POS tagging is better than TSTLA. However, TSTLA can achieve the optimal performance of WS and NER when syllables are used as the segmentation granularity. TSTLA (PRED) outperforms the other models in both WS and NER. The effect of the POS tagging task in the case of unknown word segmentation is close to that of the model with known correct word segmentation. In the WS task, the F1 values of the model in this paper are higher than those of the other three models, indicating that the method in this paper, based on ROBERTa pre-trained language model for multi-task learning implicitly performs better word segmentation by sharing data and parameters. TSTLA (PRED) compared to TSTLA (GOLD) effect in POS tagging and NER is only slightly degraded. This indicates that TSTLA (PRED) is more applicable while ensuring performance. The experimental results demonstrate the better performance of our proposed joint model for WS, POS tagging, and NER.

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5.2 | Effectiveness of Two-Stage Classification

To assess the efficacy of our proposed unified two-stage classification method, we perform the joint task of WS and POS tagging on the UD2 dataset. Additionally, we carry out the joint task of entity boundary detection and entity label classification on LST20, without employing GTT. The results are presented in Table 6.

The experiments on the UD2 dataset demonstrate that our model achieves top performance in both WS and POS tagging tasks. From the observations presented in Table 4, it is evident that BiRNN-CRF performs comparatively weaker than other joint models utilising pre-trained language models. When compared to models utilising the same pre-trained language model, TSTLA outperforms TWASP (BERT) and TWASP (ZEN), indicating the effectiveness of our two-stage classification method. Furthermore, our model achieves superior performance compared to the SPANSEGTAG model, validating the superiority of our proposed approach over previously proposed two-stage classification methods.

In the experiments conducted on the LST20 dataset, our model demonstrates superior performance in the NER task. Particularly, when comparing the Wangchanberta-syllable-CRF model and the TSTLA model, it is worth noting that both models employ the same pre-trained language model and utilise syllables as input. This ensures a fair and objective comparison between them. From the findings presented in Table 4, it is evident that the TSTLA model exhibits significantly better performance than the Wangchanberta-syllable-CRF model in terms of both macro-averaged F1 values and micro-averaged F1 values. This indicates that our proposed two-stage classification method outperforms the traditional sequence labelling method in efficiency and overall performance.

5.3 | Effects of Different Interaction Strategies

In this paper, we propose a new information interaction method named GTT, which can handle both one-way and two-way interactions. Word segmentation is the basis for POS tagging. An entity consists of one or more words, so the entity boundary must overlap with the word boundary. Entities are usually nouns or noun phrases, so POS labels can provide helpful information for entity recognition. Based on the above correlations, we propose three different interaction strategies, including M1, M2 and M3. M1 indicates one-way interaction for word boundary detection to entity boundary detection, and one-way interaction for POS classification and entity classification. M2 indicates two-way interaction for word boundary detection to entity boundary detection, and two-way interaction for POS classification and entity classification. M3 indicates a two-way interaction for word boundary detection to POS classification, a one-way interaction for word boundary detection and POS classification to entity boundary detection and one-way interaction from POS classification and entity boundary detection to entity classification. The results are reported in Table 7.

In this experiment, we find that M1 has the best overall performance and M3 has the worst performance. M1 illustrates that word boundary information contributes well to entity boundaries and POS labelling information contributes well to entity labels. The use of GTT for strongly associated tasks can promote the model performance. Comparing M1 and M2, we find that M2 works worse than M1, indicating that one-way interactions work better than two-way interactions, and that downstream task to upstream task interactions do not help the model performance. M3 has more interactions between tasks, but too many complex interactions may introduce noise to the model, which leads to performance degradation instead.

5.4 | The Effect of GTT

In order to further verify the effect of GTT on the model results we compared the models with and without GTT on the PKU dataset and the LST20 dataset to investigate the effect on model performance. The results are reported in Figure 2.

Figure 2 shows the results of the comparison experiments with the GTT (/w GTT) with and without the GTT (/wo GTT) on the PKU datasets and the LST20 datasets, we observe that on each experimental dataset, the performance of WS, POS, and NER is improved to varying degrees when using the GTT compared to not using the GTT. Among them, the macro-averaging F1 score of POS tagging and the micro-averaging F1 score of NER have very significant improvements on the PKU dataset. This suggests that the GTT approach for POS and NER tasks significantly contributes to enhancing the performance of these tasks and improving the model's generalisation capability. The performance of the WS task is also indirectly boosted due to the

TABLE 6 | Performance comparison of two-stage classification methods.

Dataset	Model	WS F1	POS Micro F1/Macro F1	NER Micro F1/Macro F1
UD2	BiRNN-CRF	95.09	89.42/-	—
	TWASP (BERT)	98.33	95.46/-	—
	TWASP (ZEN)	98.18	95.49/-	—
	SPANSEGTAG	98.12	95.54/-	—
	TSTLA	**98.59**	**95.62/93.18**	—
LST20	Wangchanberta-syllable-CRF	—	—	76.45/70.37
	TSTLA	—	—	**78.43/72.39**

Note: The bold values are the-state-of-art results.

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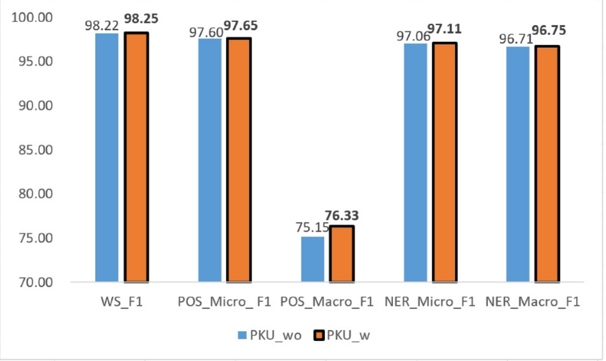
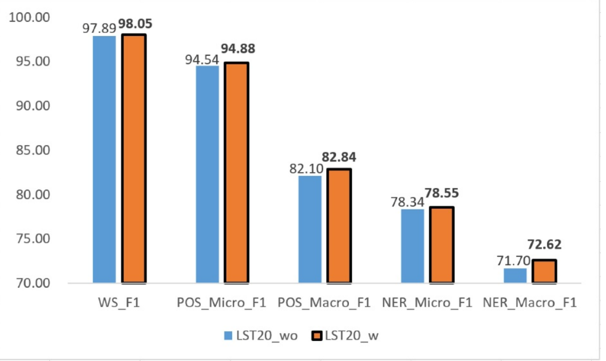
TABLE 7 | Comparative experimental results of different interaction strategies.

Dataset	Interaction strategy	WS F1	POS Micro F1/Macro F1	NER Micro F1/Macro F1
PKU	M1	**98.25**	**97.65/76.33**	**97.11/96.75**
	M2	98.22	97.53/74.54	97.10/96.73
	M3	98.22	97.55/73.82	97.08/96.74
LST20	M1	**98.05**	**94.88/82.84**	**78.55/72.62**
	M2	97.97	94.57/82.88	78.41/71.14
	M3	98.01	94.62/82.24	77.94/69.78

Note: The bold values are the-state-of-art results.



(a) Comparative experimental results of the PKU dataset



(b) Comparative experimental results of the LST20 dataset

FIGURE 2 | Comparative experimental results of with the GTT (/w GTT) with and without the GTT (/wo GTT).

improvement of POS and NER effects. The effect on the LST20 dataset is not as significant, probably because the Thai prefixes and suffixes are inherently indicative of the lexical analysis task, and thus the role of task association between POS and NER is weakened. However, we found that good performance was also observed for each task without using GTT, illustrating the effectiveness of the unified two-stage classification approach.

5.5 | Comparison of Different Hidden Size

In our model, the hidden size directly affects the performance of GTT and is therefore a vital hyperparameter. It thus requires careful consideration. In this subsection, we conducted experiments to evaluate the sensitivity of our model's performance to different hidden size settings. Specifically, we evaluated the impact of different hidden size values on the experimental results using the PKU and LST20 datasets. We experimented with hidden sizes of 32, 64, 128, 256, and 512, and the corresponding results are reported in Table 8.

From Table 8, we find that the overall performance is best for the hidden size of 64 in the PKU and LST20 datasets. In both datasets, a small hidden size tends to result in insufficient feature capture and the ignoring of available information. As the hidden size increases, the model's recognition performance peaks when the hidden size is 64 or 128, with comparable results between these two settings. When the hidden size is too large, the performance of the model starts to decrease because it will appear to be overfit. In the PKU dataset, we find that the performance of the WS and NER tasks is comparable for hidden size of 64 and 128, whereas the Macro F1 value of POS is significantly higher for hidden size of 64 than for hidden size of 128. In the LST20 dataset, although the Macro F1 value of NER is highest for hidden size of 256, all other metrics are lower than the case of hidden size of 64. Considering the overall accuracy, we set the hidden size to 64.

5.6 | Case Study

This subsection provides case studies as examples to showcase the effectiveness of GTT as presented in Table 9. We compare the prediction results under the PKU dataset with and without GTT. Comparing the experimental results, we found that the effect of with or without GTT on word and entity boundary detection was not significant, but the prediction of word and entity labels became more accurate after using GTT.

In Table 9, we give four real examples for illustration. In examples (a), we find that without GTT, the judgement of labels of nested entities is easily confused. “福建省下岗再就业基金会 (Fujian Province Layoffs and Re-employment Foundation)” is a organisation (nt), which includes “福建省 (Fujian Province)” is a location (ns) However, after using GTT, the recognition of entity labels is implicitly assisted with POS labels due to the task association between POS tagging and NER.

The labels of “麒麟” (Kirin) and “高尚 (Noble)” are ns in example (b), which are not common labels for these words.

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TABLE 8 | Comparative experimental results of different hidden size.

Dataset	Hidden size	WS F1	POS Micro F1/Macro F1	NER Micro F1/Macro F1
PKU	32	98.22	96.20/72.53	97.03/96.68
	64	**98.25**	**97.65**/**76.33**	97.11/96.75
	128	**98.25**	**97.65**/74.86	**97.13**/**96.79**
	256	98.19	96.17/72.78	97.07/96.70
	512	98.16	96.16/71.14	96.93/96.61
LST20	32	97.95	94.83/80.86	78.67/72.09
	64	**98.05**	**94.88**/**82.84**	**78.55**/72.62
	128	97.90	94.82/81.09	78.16/72.03
	256	97.87	94.82/81.33	78.43/**73.26**
	512	97.92	94.67/80.51	78.15/72.16

Note: The bold values are the-state-of-art results.

TABLE 9 | Several real cases to illustrate the effectiveness of GTT.

Example (a): The case of nested entity recognition	
/w GTT	这/r 是/v 新/d 成立/v 的/u[[福建省/ns]ns 下岗/v 再/d 就业/v 基金会/n]nt 得到/v的/u 第一/m 笔/q 捐款/n 。/w
/wo GTT	这/r 是/v 新/d 成立/v 的/u[[福建省/ns]ns 下岗/v 再/d 就业/v 基金会/n]**ns** 得到/v的/u 第一/m 笔/q 捐款/n 。/w
Example (b): The case of ambiguous words	
/w GTT	当地人/n 把/p 高山/n 上/f 的/u 坝子/n 称为/v “/w 盖/Ng ”/w 。/w[麒麟盖/ns]ns上/f 住/v 着/u[麒麟/ns]ns 、/w[莲池/ns]ns 、/w[高尚/ns]ns 3/m 个/q 村/n 311/m户/q 农民/n 。/w
/wo GTT	当地人/n 把/p 高山/n 上/f 的/u 坝子/n 称为/v “/w 盖/Ng ”/w 。/w 麒麟盖/**n** 上/f 住/v 着/u 麒麟/**nz** 、/w 莲池/**nz** 、/w 高尚/**nz** 3/m 个/q 村/n 311/m 户/q 农 民/n 。/w
Example (c): The case of unknown words	
/w GTT	(/w[孙/nr 玉坤/nr]nr[蒯/nr 创/nr]nr)/w
/wo GTT	(/w[孙/nr 玉坤/nr]nr 蒯/**Vg** 创/nr)/w

Note: *After ‘/’ is the POS tag and the entity in parentheses ‘[]’. Bold the labels that are predicted incorrectly.

Thus without GTT, these words are not tagged as entities as well as POS labels are also predicted incorrectly. Whereas with GTT part-of-speech and entity labels can be correctly predicted. This illustrates that with GTT can better incorporate contextual information. In example (c), The word “蒯 (Kuai)” is not in the BERT vocabulary, so it is tagged as [‘UNK’], which is an unknown word. Nevertheless, with GTT, the entities containing “蒯” can still be correctly predicted. This indicates that with GTT can better predict the labels of unknown words and enhance the generalisation ability of the model.

6 | Conclusion

In this paper, we propose a unified joint model for lexical analysis named TSTLA. This model handles multi-task lexical analysis problems through two-stage sequence span classification, which has a global perspective consistent with human lexical analysis intuition. In addition, we propose a Gated Task Transformation (GTT) approach that can efficiently share valuable features between tasks. Based on experiments on public datasets of Chinese and Thai, we demonstrate that TSTLA can achieve state-of-the-art results. Our future work will focus on further expanding in two directions: on the one hand, we plan to broaden the application scope of the research methods proposed in this paper to more areas of natural language processing, such as dependency parsing, semantic role labelling etc. On the other hand, we will also strive to extend these research methods to other languages, such as Vietnamese, Lao etc.

7 | Limitations

The TSTLA model proposed in this paper constructs a unified two-stage span labelling framework that ingeniously leverages the correlations among WS, POS tagging and NER to achieve joint learning of these three tasks. Since the model is performed

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based on span classification, it relatively relies on pre-trained language models [9]. Consequently, in the absence of support from pre-trained language models, the performance of our model may decline.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data will be made available on request.

Endnotes

[1] We use the PKU downloaded from https://doi.org/10.18170/DVN/SEYRX5.

[2] We use the LST20 downloaded from https://aiforthai.in.th.

[3] We use its version 2.4 downloaded from https://universaldependencies.org/.

[4] The conversation scripts are from https://github.com/skydark/nstools/tree/master/zhtools.

[5] http://sighan.cs.uchicago.edu/bakeoff2005/.

[6] https://github.com/SVAIGBA/TwASP.

[7] https://huggingface.co/bert-base-chinese.

[8] https://huggingface.co/airesearch/wangchanberta-base-att-spm-uncased.

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